

kinetic-kit

API Reference

v0.1.1

This document lists all public symbols exported by `kinetic-kit`.

1 Template

`thesis()` is the main entrypoint of the package. By default, it assembles a complete doctoral thesis in the KSP-approved style. The template is fully configurable and can also be used to produce a bachelor's or master's thesis. Note that the KSP endorsement only applies to doctoral theses in the default configuration; customized documents are not covered by it.

1.1 thesis

- `thesis()`

1.1.1 thesis

KIT thesis template — doctoral, Master's, Bachelor's and Diploma theses.

The document type is decided by the title page. `title-page` defaults to the KSP-approved doctoral page; pass your own for anything else.

1.1.1.1 Parameters

```
thesis(  
  author-firstname: str,  
  author-surname: str,  
  title: content,  
  title-page,  
  format,  
  lang,  
  margin-preset,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  abstract-en,  
  abstract-de,  
  acknowledgements,  
  notation,  
  abbreviations,  
  show-list-of-figures: bool,  
  show-list-of-tables: bool,  
  show-list-of-listings: bool,  
  figure-kinds: array,  
  appendix,  
  bibliography,  
  own-publications,  
  own-patents,  
  supervised-theses,  
  doc: content  
) -> content
```

1.1.1.1.1 author-firstname str

Author's first name.

Default: "Max"

1.1.1.1.2 author-surname str

Author's surname.

Default: "Mustermann"

1.1.1.1.3 title content

Thesis title.

- title-page (content | function | none): Title page. Defaults to the doctoral page; pass doctoral-title-page.with(...) to configure it, your own content, a function called as (title, author-firstname:, author-surname:, format:, lang:), or none to omit it.
- format ("a5" | "17x24" | "a4"): Paper format — "a5" (148×210 mm, default), "17x24" (170×240 mm), or "a4" (210×297 mm, discouraged by KSP). Font sizes and margins are set automatically.
- lang ("de" | "en"): Document language.
- margin-preset ("short" | "medium" | "long"): Margin profile keyed on page count.

Default: [Your Thesis Title]

1.1.1.1.4 binding-correction length

BCOR added to inside margin. Default 0mm.

Default: 0mm

1.1.1.1.5 colored-links bool

KIT Blue links when true, black when false.

Default: true

1.1.1.1.6 draft bool

Show "ENTWURF" watermark when true.

- draft-info (str | none): Optional version string below watermark. Default none.

Default: false

1.1.1.1.7 serif-headings bool

Use serif font for headings when true. Default false (sans-serif).

Default: false

1.1.1.1.8 heading-numbering-depth int

Deepest heading level to number. Default 3.

- abstract-en (content | none): English abstract. none = omit.
- abstract-de (content | none): German abstract. none = omit.
- acknowledgements (content | none): Acknowledgements. none = omit.
- notation (content | none): Notation list. none = omit.
- abbreviations (content | none): Abbreviations list. none = omit.

Default: 3

1.1.1.1.9 show-list-of-figures bool

Include List of Figures.

Default: `true`

1.1.1.1.10 show-list-of-tables bool

Include List of Tables.

Default: `true`

1.1.1.1.11 show-list-of-listings bool

Include List of Listings.

Default: `false`

1.1.1.1.12 figure-kinds array

Figure kinds beyond image, table and raw, as dictionaries with kind, supplement, and optionally list-title and show-list. supplement and list-title take either one value or one per language, e.g. (de: [Algorithmus], en: [Algorithm]). The built-in kinds are not declared here — their list pages are governed by show-list-of-figures / show-list-of-tables / show-list-of-listings. Declared kinds get a list page after the built-in ones, in declaration order.

- appendix (content | none): Appendix chapters. Template applies A, A.1, ... numbering and places the appendix before the back-matter lists. none = omit.
- bibliography (content | none): Bibliography content. Pass bibliography("refs.bib", title: none, style: "ieee"). The template adds a translated heading. none = omit.
- own-publications (content | none): Own publications content (heading added by template). none = omit.
- own-patents (content | none): Own patents content (heading added by template). none = omit.
- supervised-theses (content | none): Supervised theses content (heading added by template). none = omit.

Default: ()

1.1.1.1.13 doc content

Main document body (chapters only).

2 Style Constants

The kit-style dictionary exposes the template's visual constants so that custom figures, diagrams, and other content can match the document's typography and KIT color palette exactly.

```
#import "@preview/kinetic-kit:0.1.1": kit-style

#let font-sizes = kit-style.font-sizes-by-format.at("a5")
#set text(font: kit-style.fonts.sans, size: font-sizes.small)
#rect(fill: kit-style.colors.green15, stroke: kit-style.colors.green)
```

2.1 Fonts

Field	Value	Description
fonts.serif	("Libertinus Serif",)	Serif body text
fonts.sans	("Libertinus Sans",)	Sans-serif headings and headers
fonts.mono	("Libertinus Mono",)	Code listings
leading	0.75em	Paragraph line spacing ($\approx 1.15\times$ at 10 pt)

2.2 Font Sizes

font-sizes-by-format is a dictionary keyed by format string ("a5", "17x24", "a4"). Each entry contains the sizes below. Access them with `kit-style.font-sizes-by-format.at(format)`.

All fields are length. Example values shown for the "a5" format (10 pt base).

Field	A5 / 17x24	A4	Description
base	10pt	11pt	Body text
chapter	18pt	25pt	Chapter heading (level 1)
section	14pt	17pt	Section heading (level 2)
subsection	12pt	14pt	Subsection heading (level 3)
subsubsection	10pt	11pt	Subsubsection heading (level 4+)
small	8pt	9pt	Running headers, footnotes, captions
footnote	8pt	9pt	Footnote text
title	18pt	25pt	Title page — document title
author	14pt	14pt	Title page — author name
title-info	14pt	14pt	Title page — info lines

2.3 Colors

All values are color. KIT brand colors follow a systematic naming scheme: base name plus optional opacity suffix (70, 50, 30, 15 = 70 %, 50 %, 30 %, 15 % tint).

Field	Hex	Description
colors.green	#009682	KIT Green — primary brand color
colors.green70	#4CB5A7	
colors.green50	#7FCAC0	
colors.green30	#B2DFD9	
colors.green15	#D9EFEC	
colors.blue	#4664AA	KIT Blue — links and accents
colors.blue70	#7D92C3	
colors.blue50	#A2B1D4	
colors.blue30	#C7D0E5	
colors.blue15	#E3E8F2	
colors.black	#000000	
colors.black70	#4D4D4D	
colors.black50	#808080	
colors.black30	#B3B3B3	

Field	Hex	Description
colors.black15	#D9D9D9	
colors.palegreen	#82BE3C	KIT extended palette
colors.yellow	#FAE614	
colors.orange	#DCA01E	
colors.brown	#A08232	
colors.red	#A01E28	
colors.lilac	#A00078	
colors.cyanblue	#50AAE6	
colors.keyword	#0000C8	Syntax highlighting — keyword
colors.comment	#3F7F5F	Syntax highlighting — comment
colors.string	#700055	Syntax highlighting — string literal

3 Components

The components module exports the individual building blocks for assembling a document without the full thesis() orchestrator. All symbols are accessed through the components namespace:

```
#import "@preview/kinetic-kit:0.1.1": components
```

```
#let format = "a5"
```

```
#show: components.setup-page.with(format: format, margin-preset: "short", lang: "de")
```

```
#show: components.setup-front-matter
```

```
#components.table-of-contents(lang: "de")
```

You are responsible for applying the wrappers in the correct order: setup-page → setup-front-matter → content → setup-content → (optionally) setup-appendix.

3.1 Page Setup

3.2 page-setup

- setup-page()
- setup-front-matter()
- setup-content()
- setup-appendix()

3.2.1 setup-page

Apply the full KIT document style: page geometry, running headers, KSP typography, heading styles, figure captions, equations, and code blocks. Use as a show rule: #show: setup-page.with(...).

3.2.1.1 Parameters

```
setup-page(  
  format: str,  
  margin-preset: str,  
  lang: str,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info: content,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  figure-kinds: array,  
  doc: content  
) -> content
```

3.2.1.1.1 format str

Paper format — "a5" (148×210 mm, default), "17x24" (170×240 mm), or "a4" (210×297 mm).
Font sizes and margins are set automatically per KSP specifications.

Default: "a5"

3.2.1.1.2 margin-preset str

Margin profile keyed on expected page count — "short" (under 200 pp), "medium" (200–399 pp),
"long" (400+ pp).

Default: "short"

3.2.1.1.3 lang str

Document language — "de" or "en".

Default: "de"

3.2.1.1.4 binding-correction length

Extra inside margin added for binding (e.g. 3mm).

Default: 0mm

3.2.1.1.5 colored-links bool

Render external hyperlinks in KIT blue when true.

Default: true

3.2.1.1.6 draft bool

Show the draft watermark on every page when true.

Default: false

3.2.1.1.7 draft-info `content`

Optional extra text appended to the watermark (e.g. a git SHA).

Default: `none`

3.2.1.1.8 serif-headings `bool`

Use Libertinus Serif for headings when true. Default false

Default: `false`

3.2.1.1.9 heading-numbering-depth `int`

Deepest heading level that receives a number. Default 3. Headings deeper than this are styled normally but rendered without a number or indent grid.

Default: `3`

3.2.1.1.10 figure-kinds `array`

Figure kind declarations, merged onto the built-in `image`, `table` and `raw` entries. Supplies the caption supplements; printing the matching list pages is the caller's job.

Default: `()`

3.2.1.1.11 doc `content`

Document body (injected automatically by the show rule).

3.2.2 setup-front-matter

Switch to Roman numeral page numbering and remove heading numbering. Apply before front-matter content: `#show: setup-front-matter.`

3.2.2.1 Parameters

`setup-front-matter`(doc: `content`) -> `content`

3.2.2.1.1 doc `content`

Document body (injected automatically by the show rule).

3.2.3 setup-content

Switch to Arabic page numbering and enable 1.1 heading numbering. Apply before the main content: `#show: setup-content.`

3.2.3.1 Parameters

```
setup-content(doc: content) -> content
```

3.2.3.1.1 doc content

Document body (injected automatically by the show rule).

3.2.4 setup-appendix

Switch to A.1 heading numbering and reset the heading counter. Apply before appendix chapters:
#show: setup-appendix.

3.2.4.1 Parameters

```
setup-appendix(doc: content) -> content
```

3.2.4.1.1 doc content

Document body (injected automatically by the show rule).

3.3 Title Page

3.4 title-page

- doctoral-title-page()

3.4.1 doctoral-title-page

Render the KIT doctoral thesis title page (German legal format). All text on this page is always in German regardless of document language.

3.4.1.1 Parameters

```
doctoral-title-page(  
  title: content,  
  author-title: content,  
  author-firstname: str,  
  author-surname: str,  
  author-male: bool,  
  doc-degree: str,  
  doc-degree-f: str,  
  department: str,  
  university-genitive: str,  
  status-approved: bool,  
  exam-date: content,  
  main-advisor: content,  
  main-advisor-male: bool,  
  co-advisor: content,  
  co-advisor-male: bool,  
  format: str,  
  lang: str  
) -> content
```

3.4.1.1.1 title content

Dissertation title.

3.4.1.1.2 author-title content

Academic title preceding the author's name (none = omit).

Default: "M.Sc."

3.4.1.1.3 author-firstname str

Author's first name.

Default: "Max"

3.4.1.1.4 author-surname str

Author's surname.

Default: "Mustermann"

3.4.1.1.5 author-male bool

Selects grammatical gender for the degree article.

Default: true

3.4.1.1.6 doc-degree str

Degree name in masculine form (e.g. "Doktor-Ingenieur").

Default: "Doktors der Ingenieurwissenschaften (Dr.-Ing.)"

3.4.1.1.7 doc-degree-f str

Degree name in feminine form (e.g. "Doktor-Ingenieurin").

Default: "Doktorin der Ingenieurwissenschaften (Dr.-Ing.)"

3.4.1.1.8 department str

KIT department or faculty.

Default: "KIT-Fakultät für Maschinenbau"

3.4.1.1.9 university-genitive str

University name in genitive (e.g. "des Karlsruher Instituts...").

Default: "des Karlsruher Instituts für Technologie (KIT)"

3.4.1.1.10 status-approved `bool`

false = submitted version, true = approved version.

Default: `false`

3.4.1.1.11 exam-date `content`

Date of oral examination — shown only when status-approved is true.

Default: `none`

3.4.1.1.12 main-advisor `content`

Main referee — shown only when status-approved is true.

Default: `none`

3.4.1.1.13 main-advisor-male `bool`

Selects gendered label for the main advisor.

Default: `true`

3.4.1.1.14 co-advisor `content`

Co-referee — shown only when status-approved is true.

Default: `none`

3.4.1.1.15 co-advisor-male `bool`

Selects gendered label for the co-advisor.

Default: `true`

3.4.1.1.16 format `str`

Paper format — "a5", "17x24", or "a4". Determines font sizes and title-page margins.

Default: `"a5"`

3.4.1.1.17 lang `str`

Accepted for the title-page contract and ignored — this page is always German.

title, author-firstname, author-surname, format and lang are supplied by the template when this is passed to title-page; setting them via `.with()` has no effect there.

Default: `"de"`

3.5 Outlines

3.6 outlines

- `setup-outlines()`
- `table-of-contents()`
- `list-of()`
- `list-of-figures()`
- `list-of-tables()`
- `list-of-listings()`

3.6.1 setup-outlines

Shared outline styling, applied as a show rule so every outline in the document — including a user's own — matches.

3.6.1.1 Parameters

```
setup-outlines(body: content) -> content
```

3.6.1.1.1 body content

Document body (injected automatically by the show rule).

3.6.2 table-of-contents

Print the table of contents.

3.6.2.1 Parameters

```
table-of-contents(lang: str) -> content
```

3.6.2.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.6.3 list-of

Print a back-matter list page for one figure kind.

Handles the in-outline state that switches flex-caption to its short form, which a hand-rolled `outline(target: ...)` would miss.

- `kind` (function | str): Figure kind to list — an element function such as `image`, or a string such as "algorithm".
- `title` (content | str): Heading of the list page.

3.6.3.1 Parameters

```
list-of(  
  kind,  
  title  
) -> content
```

3.6.4 list-of-figures

Print the list of figures.

3.6.4.1 Parameters

```
list-of-figures(lang: str) -> content
```

3.6.4.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.6.5 list-of-tables

Print the list of tables.

3.6.5.1 Parameters

```
list-of-tables(lang: str) -> content
```

3.6.5.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.6.6 list-of-listings

Print the list of listings.

3.6.6.1 Parameters

```
list-of-listings(lang: str) -> content
```

3.6.6.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.7 Figures

3.8 figures

- flex-caption()
- setup-figures()

3.8.1 flex-caption

Two-part caption: short version for LoF/LoT, long version under the figure.

Usage: `#figure(..., caption: flex-caption(short: [Short], long: [Long.] )`

3.8.1.1 Parameters

```
flex-caption(  
  short: content,  
  long: content  
) -> content
```

3.8.1.1.1 short content

Short caption shown in List of Figures / Tables.

Default: none

3.8.1.1.2 long content

Full caption shown below the figure in the document body.

Default: none

3.8.2 setup-figures

Shared figure, caption, and table styling. Apply as a show rule.

3.8.2.1 Parameters

```
setup-figures(  
  font-sizes: dict,  
  lang: str,  
  figure-kinds: array,  
  body: content  
) -> content
```

3.8.2.1.1 font-sizes dict

Format-specific font sizes resolved by the template.

3.8.2.1.2 lang str

Document language — "de" or "en".

Default: "de"

3.8.2.1.3 figure-kinds array

Figure kind declarations, merged onto the built-in ones. Only the supplements are read here; list pages are printed by the template's back matter.

Default: ()

3.8.2.1.4 body content

Document body (injected automatically by the show rule).

3.9 Figure Kinds

Typst keeps a separate counter and supplement for every figure kind. The template carries strings for Typst's own image, table and raw, because those names are template chrome; their list pages are governed by the `show-lo*` booleans. Any other kind — pseudocode, theorems, whatever a document needs — is declared through the `figure-kinds` parameter of `thesis()`, which supplies the caption supplement and, optionally, a back-matter list page.

Per-chapter counter resets are not part of this registry. `setup-headings` derives them from the figures present in the document, so every kind restarts each chapter whether or not it has been declared.

```
#show: thesis.with(  
  figure-kinds: (  
    (  
      kind: "algorithm",  
      supplement: (de: [Algorithmus], en: [Algorithm]),  
      list-title: (de: [Algorithmenverzeichnis], en: [List of Algorithms]),  
      show-list: true,  
    ),  
    // Supplement only — no list page.  
    (kind: "theorem", supplement: (de: [Satz], en: [Theorem])),  
  ),  
)
```

3.10 figure-kinds

- `resolve-localized()`
- `resolve-figure-kinds()`

3.10.1 resolve-localized

Pick the variant of a per-language value that matches a language.

Values may be plain content or a dictionary keyed by language code. A plain value is used for every language; a dictionary is looked up by `lang`, then by `fallback`, and fails loudly if neither is present.

- `value` (`content` | `str` | `dict`): Plain value, or one variant per language code.

3.10.1.1 Parameters

```
resolve-localized(  
  value,  
  lang: str,  
  fallback,  
  kind,  
  field: str  
) -> content | str
```

3.10.1.1.1 lang str

Language to look up first — usually `text.lang` at the use site.

- fallback (str | none): Language to fall back to, usually the document language.
- kind (function | str | none): Figure kind, used in the error message only.

3.10.1.1.2 field str

Field name, used in the error message only.

Default: `"value"`

3.10.2 resolve-figure-kinds

Resolve the document's figure kinds into one ordered list.

The built-in kinds come first, in the canonical order of their list pages, with visibility taken from the three booleans. Kinds the document declares follow, in declaration order. Declaring a built-in kind is an error.

3.10.2.1 Parameters

```
resolve-figure-kinds(  
  figure-kinds: array,  
  show-list-of-figures: bool,  
  show-list-of-tables: bool,  
  show-list-of-listings: bool  
) -> array
```

3.10.2.1.1 figure-kinds array

Entries declared by the document.

3.10.2.1.2 show-list-of-figures bool

Whether the list of figures is printed.

Default: `true`

3.10.2.1.3 show-list-of-tables bool

Whether the list of tables is printed.

Default: `true`

3.10.2.1.4 show-list-of-listings bool

Whether the list of listings is printed.

Default: `false`

3.11 Headings

3.12 headings

- setup-headings()

3.12.1 setup-headings

Shared heading styling: per-level sizes and spacing, chapter page breaks and counter resets, and number-body alignment. Apply as a show rule.

3.12.1.1 Parameters

```
setup-headings(  
  font-sizes: dict,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  body: content  
) -> content
```

3.12.1.1.1 font-sizes dict

Format-specific font sizes resolved by the template.

3.12.1.1.2 serif-headings bool

Use Libertinus Serif for headings when true.

Default: false

3.12.1.1.3 heading-numbering-depth int

Deepest heading level that receives a number.

Default: 3

3.12.1.1.4 body content

Document body (injected automatically by the show rule).